

Accurate Source-Free Speech Classification via Meta-Learned Target-Centric Model Merging

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Abstract

How can we build a speech classifier for an unseen domain when access to source-domain data is unavailable? Speech classification is widely used (e.g., voice commands, emotion recognition, etc.), but it is highly sensitive to domain shifts (e.g., languages), often causing substantial performance degradation. This issue becomes critical when source data cannot be accessed and only pretrained source models with limited target data are available. Under such constraints, typical adaptation approaches are inapplicable, and existing model merging methods easily overfit the limited target data. We propose MOCHEE, an accurate target-centric model merging framework that combines multiple pretrained source models using limited target data. To improve generalization under scarce target labels, MOCHEE exploits meta-learning to resolve representation misalignment and emphasize target-relevant sources. Extensive experiments show that MOCHEE outperforms the baselines by up to 14.5 points in Macro-F1.

Index Terms: speech classification, model merging, source-free, target-centric, meta-learning, meta-reweighting

1. Introduction

Given multiple pretrained source-domain speech classifiers but no access to the original source data, how can we build an accurate classifier for an unseen target domain using only limited labeled target data? Speech classification assigns labels to acoustic signals and supports many real-world systems, including emotion recognition [1, 2, 3], speaker identification [4, 5], language identification [6, 7], and fake speech detection [8, 9]. We focus on *source-free* setting where only multiple pretrained source classifiers and a small labeled target set are available. This setting is common in practice because source speech data are often private or proprietary, whereas pretrained models can still be shared for downstream use. For example, hospitals and call centers release pretrained models while keeping raw recordings as confidential assets.

Under this constraint, many transfer learning [10, 11, 12], data selection [13, 14, 15], and meta-learning [16, 17] approaches become difficult to apply, as they typically require access to source-domain data. Recently, model merging [18, 19, 20] has emerged as a promising alternative under such source-free constraints. Model merging consolidates multiple pretrained models into a single model via averaging or heuristic alignment strategies, without updating source parameters. While some approaches [18, 21] use a held-out set, this set is primarily used for model selection rather than learning domain-specific relevance. Moreover, when only a few labeled target

Table 1: Comparison of representative approaches for adapting speech classifiers to a new target domain. We summarize whether each category supports source-free deployment, fast adaptation, and robustness in low-shot target regimes.

Method	Source-free	Fast adaptation	Target generalization
Transfer learning			✓
Meta learning		✓	✓
Model merging	✓	✓	
MOCHEE (proposed)	✓	✓	✓

examples are available, such selection or combination decisions become highly sensitive to the limited supervision, leading to biased selection and unstable target performance. We summarize these properties and limitations in Table 1.

To address the limitations of existing model merging methods for source-free speech classification, we propose MOCHEE (TARGET-AWARE MODEL MERGING FOR SPEECH CLASSIFICATION IN A SOURCE-FREE SETTING), an accurate target-centric method that learns generalization-oriented merging scores for each source model. Unlike prior works that optimize merging weights to perform well on the available target data, we learn target-relevance scores for each source model with a meta-learning-based objective to promote generalization to unseen target samples. Moreover, since independently trained source models can be misaligned by hidden-unit permutations, MOCHEE aligns them before merging. This design ensures that the learned merging scores do not overfit to the limited labeled target instances, but instead capture generalization-oriented source contributions that transfer to unseen target data.

Our main contributions are as follows. First, we highlight an under-studied yet practical *source-free* setting for speech classification. Second, we propose a generalization-oriented model merging method that learns source importance via a meta-learning objective and performs feature-permutation alignment to reliably merge misaligned source models. Third, on cross-corpus speech classification benchmarks, MOCHEE consistently outperforms strong baselines, demonstrating improved target-domain generalization under limited target supervision. Our code is publicly released at <https://github.com/snudatalab/Mochee>.

2. Related Work

2.1. Model Merging

Model merging combines pretrained models without retraining. Model soups [18] averages learned parameters but often fails under domain heterogeneity due to negative transfer. Permutation-aware merging [19] mitigates neuron-level misalignment via hard, layer-wise matching, but it can be brittle

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when the alignment signal is weak. In speech, model merging is widely used in automatic speech recognition (ASR) [22, 23, 24, 25], where checkpoint/weight averaging is a common recipe that improves generalization with negligible overhead. Model merging has also been applied to disordered speech ASR [26]. Beyond ASR, averaged pretrained checkpoints are used in speech emotion recognition and other speech classification pipelines [27]. However, they use model merging primarily as within-domain checkpoint averaging to gain ensemble-like robustness. Domain and data heterogeneity have been studied across diverse areas, such as graph learning [28, 29, 30, 31, 32] and semi-supervised or domain-aware speech learning [33, 34]. In our source-free setting with heterogeneous source models and scarce target instances, existing methods often select models by directly minimizing the target training loss [35, 36] or by heuristic interference resolution [37], which can be unstable and overfit to the limited target supervision. Beyond model merging, efficient representation and factorization methods have also been developed for heterogeneous or evolving data, including spectral transforms for time series [38, 39], compact multi-way factorization [40, 41, 42, 43, 44], streaming and irregular tensor decomposition [45, 46, 47, 48, 49, 50], and structured sequence understanding [51]. In contrast, MOCHEE focuses on source-free speech adaptation by aligning and merging pretrained classifiers without accessing source-domain data.

2.2. Meta-reweighting

Meta-reweighting [52] improves robustness by reweighting noisy training examples using a reliable validation set. The main idea is to downweight noisy training examples and up-weight clean ones based on how well each instance’s training gradient aligns with the validation gradient.

Given a model f_θ with parameters θ and a per-instance training loss $l(x_i; \theta)$, meta-reweighting assigns a learnable score ϵ_i to each training instance: $\mathcal{L}_{\text{trn}} = \sum_{i=1}^M \epsilon_i l(x_i; \theta)$ where M is the number of training instances. It then performs a one-step virtual update $\theta(\epsilon) = \theta - \eta \nabla_\theta \mathcal{L}_{\text{trn}}$ and computes each ϵ_i from the meta-gradient of the validation loss:

$$\epsilon_i \leftarrow \max \left(0, -\eta \frac{\partial}{\partial \epsilon_i} \frac{1}{M_t} \sum_{j=1}^{M_t} l(x_j; \theta(\epsilon)) \right) \Big|_{\epsilon_i=0}, \quad (1)$$

where M_t is the number of validation instances and η is the learning rate. Intuitively, meta-reweighting learns which training examples are trustworthy teachers: it upweights examples whose training-step gradients improve validation performance, rather than fitting weights to validation loss directly.

Meta-reweighting has been widely applied to various areas including graph classification [53], noisy relation extraction [54], named entity recognition [55], and language modeling under noisy labels [56]. In speech, DoReMe [57] uses meta-reweighting for domain-aware source instance selection in target-domain speech classification. To the best of our knowledge, our proposed MOCHEE is the first to meta-reweight source-model importance for source-free speech classification.

2.3. Problem Definition

Problem 1. (Source-free Model Merging for Speech Classification.) Let the source domains be $S_1, \dots, S_N$. For each source domain S_n , we are given a pretrained speech classifier, but we do not have access to any source-domain data. At test time, we are given an unseen target domain T together with a small labeled target set $\mathcal{D}_T$ with limited labeled instances. The goal is

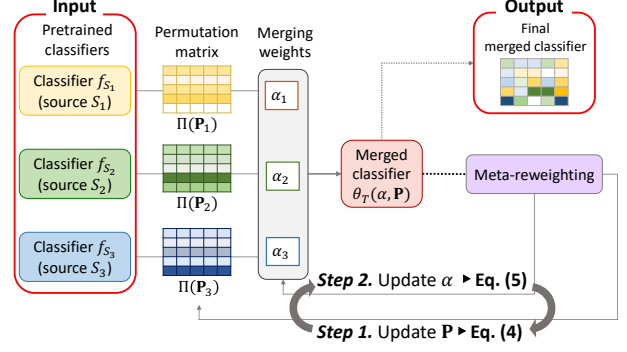


Figure 1: Overview of MOCHEE. Given pretrained source classifiers, MOCHEE learns 1) soft permutations $\mathbf{P}$ to align hidden units, and 2) merging weights α to construct a target-centric classifier $\theta_T(\alpha, \mathbf{P})$ via meta-reweighting.

to construct a target-centric speech classifier by learning how to merge the pretrained classifiers from $S_1, \dots, S_N$ using the limited supervision in $\mathcal{D}_T$, thereby maximizing generalization performance on unseen test samples from the target domain T .

3. Proposed Method

We propose MOCHEE, an accurate target-centric model merging framework for source-free speech classification. Given pretrained source-domain classifiers and a small labeled target set, MOCHEE constructs a single target-centric classifier by aligning hidden representation space and learning generalization-oriented merging weights, without updating any source-model parameters. Figure 1 illustrates the overall pipeline. We next highlight the core challenges and our corresponding ideas:

1. **How can we merge independently trained source models that are misaligned in parameter space?** Even with identical architectures, hidden units can be permuted across source models, making naive weight averaging unreliable. To address this, MOCHEE learns feature permutations to align source models before merging (Section 3.1).
2. **After aligning the parameter space, how can we learn the generalization-oriented merging scores?** We optimize the per-source merging scores with meta-reweighting, which directly promotes target generalization rather than merely minimizing the target training loss (Section 3.2).

3.1. Permutation-based Source Model Alignment

A speech classifier consists of a frozen speech embedding model (e.g., wav2vec2.0 [58] or Whisper [59]) followed by lightweight MLP heads. We focus on aligning the MLP heads, since the embedding model provides a shared representation space while the MLP head captures domain-specific decision boundaries. A challenge is that merging independently trained heads is non-trivial due to *permutation symmetry*. Even when all source MLP heads have identical architectures, independently trained heads may represent the same function with different hidden-unit indexings; permuting the hidden units of a layer and applying the corresponding inverse permutation to the next layer leaves the end-to-end mapping unchanged. As a result, the same neuron index can correspond to different semantics across sources, and naive parameter averaging mixes mismatched units. To address this, MOCHEE learns *soft permutations* that align hidden units of each source head in a target-

aware manner. Specifically, it learns layer-wise soft re-indexing matrices which map each source head to a shared hidden-unit ordering, enabling meaningful parameter merging across independently trained heads.

Let the K -layer MLP head of source S_n be parameterized as $\theta_n = \{\theta_n^{(k)}\}_{k=1}^K$ with $\theta_n^{(k)} = \{\mathbf{W}_n^{(k)}, \mathbf{b}_n^{(k)}\}$, where $\mathbf{W}_n^{(k)} \in \mathbb{R}^{H_k \times H_{k-1}}$, $\mathbf{b}_n^{(k)} \in \mathbb{R}^{H_k}$, and H_k is the output width of layer k . For each source n and layer $k \in \{1, \dots, K-1\}$, we introduce learnable $\mathbf{P}_n^{(k)} \in \mathbb{R}^{H_k \times H_k}$ and obtain a soft permutation matrix $\Pi_n^{(k)} = \Pi(\mathbf{P}_n^{(k)}) \in \mathbb{R}^{H_k \times H_k}$ via Sinkhorn normalization [60, 61]. Sinkhorn iteratively normalizes the rows and columns of $\mathbf{P}_n^{(k)}$, producing a nonnegative matrix whose row and column sums are approximately one. This doubly-stochastic constraint provides a differentiable relaxation of hard permutation matrices, encouraging near one-to-one re-indexing and thus stabilizing alignment.

Using $\{\Pi_n^{(k)}\}_{k=1}^{K-1}$, we define the aligned head parameters $\bar{\theta}_n = \{\bar{\theta}_n^{(k)}\}_{k=1}^K$, where $\bar{\theta}_n^{(k)} = \{\bar{\mathbf{W}}_n^{(k)}, \bar{\mathbf{b}}_n^{(k)}\}$ are given by

$$\bar{\mathbf{W}}_n^{(k)} = \Pi_n^{(k)} \mathbf{W}_n^{(k)} (\Pi_n^{(k-1)})^\top, \quad \bar{\mathbf{b}}_n^{(k)} = \Pi_n^{(k)} \mathbf{b}_n^{(k)}. \quad (2)$$

The right-multiplication by $(\Pi_n^{(k-1)})^\top$ applies the corresponding inverse re-indexing on the input axis of layer k (i.e., it cancels the permutation introduced at layer $k-1$) and left-multiplication by $\Pi_n^{(k)}$ softly re-indexes the output hidden units of layer k into a shared ordering across sources. We set $\Pi_n^{(0)} = \mathbf{I}_{H_0}$ (no re-indexing on the input features) and $\Pi_n^{(K)} = \mathbf{I}_{H_K}$ (no re-indexing on the final output logits). As a result, the aligned head $\bar{\theta}_n$ computes the same function as θ_n , while expressing its hidden units in a common ordering that enables meaningful parameter merging across sources.

3.2. Target-centric Merging Weights for Source Domains

Given aligned source heads $\{\bar{\theta}_n\}_{n=1}^N$ in Equation (2), our goal is to learn target-centric merging weights $\alpha = \{\alpha_n\}_{n=1}^N$ that combine the source classifiers into a single target classifier. Specifically, we construct the target head $\theta_T = \{\theta_T^{(k)}\}_{k=1}^K$ by weighted averaging in the aligned parameter space:

$$\theta_T^{(k)} = \sum_{n=1}^N \alpha_n \bar{\theta}_n^{(k)}. \quad (3)$$

A key challenge is that the labeled target set $\mathcal{D}_T$, which provides the only supervision for learning α , is extremely limited. Directly optimizing α to minimize the target training loss therefore risks overfitting to noisy target patterns.

Rather than selecting α to fit the small target training split, we formulate merging-weight learning as a meta-learning problem, where α is optimized to improve the validation performance on held-out target data. To this end, we first split $\mathcal{D}_T$ into $\mathcal{D}_T^{\text{trn}}$ and $\mathcal{D}_T^{\text{val}}$. Inspired by the meta-reweighting principle [52], we then adopt a two-stage meta-learning procedure: a virtual *inner adaptation* on $\mathcal{D}_T^{\text{trn}}$ followed by a validation-based *outer update* on $\mathcal{D}_T^{\text{val}}$.

In standard meta-learning, the inner step updates the merged model parameters θ_T . However, directly adapting θ_T on a few labeled target samples undermines the robustness and efficiency advantages of model merging. Instead, we perform the inner adaptation on the alignment parameters $p = \{\mathbf{P}_n^{(k)}\}$. The adapted permutations p' define a target-aligned merged head $\theta_T(\alpha, p')$, which is then evaluated on $\mathcal{D}_T^{\text{val}}$ to compute the meta-gradient for α .

Formally, the inner adaptation updates only the alignment

Table 2: *Dataset statistics from the CAMEO collection. Each dataset is treated as an independent domain.*

Dataset	Language	# of instances	Role
CREMA-D ¹ [62]	English	7,442	Source
SubESCO ¹ [63]	Bengali	6,000	Source
RAVDESS ¹ [64]	English	1,056	Source
MESD ¹ [65]	Spanish	862	Source
CAFÉ ¹ [66]	Canadian French	792	Target
Oréau ¹ [67]	French	430	Target

¹<https://huggingface.co/datasets/amu-cai/CAMEO>

parameters:

$$p' = p - \eta \nabla_p \left(\frac{1}{|\mathcal{D}_T^{\text{trn}}|} \sum_{d_i \in \mathcal{D}_T^{\text{trn}}} l(d_i; \theta_T(\alpha, p)) \right). \quad (4)$$

Then, following Equation (1), we compute unnormalized merging scores using the held-out split $\mathcal{D}_T^{\text{val}}$:

$$\alpha_n \leftarrow \max \left(0, -\gamma \frac{\partial}{\partial \alpha_n} \frac{1}{|\mathcal{D}_T^{\text{val}}|} \sum_{d_j \in \mathcal{D}_T^{\text{val}}} l(d_j; \theta_T(\alpha, p')) \right) \Bigg|_{\alpha_n=0}, \quad (5)$$

where γ is the step size. Since Equation (5) yields unnormalized scores, we apply a softmax normalization ($\alpha_n \leftarrow \exp(\alpha_n) / \sum_{j=1}^N \exp(\alpha_j)$) and use them as the actual merging weights in Equation (3).

Although inspired by [52], our formulation differs fundamentally in both goal and scope. While [52] performs instance-wise reweighting to suppress noisy training samples, we learn component-wise merging scores that control cross-domain model composition. Our meta-objective therefore targets robust source aggregation under scarce target supervision rather than noisy-instance filtering.

4. Experiments

Through experiments, we answer the following questions:

- Q1. Performance (Section 4.2).** How accurately does MOCHEE classify target-domain instances using only pre-trained source models and limited target supervision?
- Q2. Ablation study (Section 4.3).** Does each component of MOCHEE contribute to target-domain performance?
- Q3. Source contribution analysis (Section 4.4).** How do the learned source weights α change during training, and do they upweight the most target-relevant source domains?

4.1. Experimental Settings

Datasets. We use six speech-emotion datasets from the CAMEO collection [68], each recorded in a different language. CREMA-D [62], SubESCO [63], RAVDESS [64], and MESD [65] serve as source domains, while CAFÉ [66] and Oréau [67] serve as target domains. We map all datasets to six common emotion classes (anger, disgust, fear, happiness, neutral, and sadness), discard other labels, and report the resulting statistics in Table 2. Source datasets are split by 8 : 1 : 1 into train/validation/test sets for source-model pretraining. Target datasets are split by 1 : 9 and 2 : 8 into training/test sets for CAFÉ and Oréau, respectively; for MOCHEE, each target training set is split equally into target-training and target-validation sets.

Baselines. We compare MOCHEE against existing model merging techniques. *Uniform* naively averages source classifier heads and *Greedy Soup* [18] incrementally selects models based on validation performance. *Re-basin* [19] aligns hidden units

Table 3: Classification performance on target domains (CAFÉ and Oréau). Best results are in bold and second-best results are underlined. MOCHEE consistently shows the best performance.

Target	CAFÉ		Oréau	
Method	Accuracy	Macro-F1	Accuracy	Macro-F1
Uniform	36.78 $\pm$ 0.73	26.43 $\pm$ 0.61	22.55 $\pm$ 1.35	14.84 $\pm$ 1.10
Re-basin + Uni.	43.44 $\pm$ 0.90	39.68 $\pm$ 1.16	22.61 $\pm$ 1.75	16.65 $\pm$ 1.52
Greedy Soup	41.45 $\pm$ 1.09	42.01 $\pm$ 1.25	25.04 $\pm$ 1.49	20.38 $\pm$ 1.80
TIES	38.35 $\pm$ 0.52	30.02 $\pm$ 0.67	23.07 $\pm$ 1.15	17.55 $\pm$ 1.79
Re-basin + TIES	42.66 $\pm$ 0.62	36.53 $\pm$ 0.55	23.48 $\pm$ 1.63	17.57 $\pm$ 1.37
MOCHEE	50.85 $\pm$ 1.74	49.40 $\pm$ 1.94	35.30 $\pm$ 1.74	34.85 $\pm$ 2.17

Table 4: Tuned performance of source-pretrained models and merging baselines on CAFÉ. FT denotes fine-tuning on the labeled target training set. All methods except MOCHEE are fine-tuned. Best results are in bold. MOCHEE achieves the highest accuracy without any additional fine-tuning, outperforming even baselines that are fine-tuned on the target data.

Category	Method	Accuracy	Macro-F1
Single-source+FT	CREMA-D	44.59 $\pm$ 0.74	44.77 $\pm$ 1.21
	SubESCO	45.82 $\pm$ 0.58	46.41 $\pm$ 0.44
	RAVDESS	45.90 $\pm$ 2.15	46.01 $\pm$ 2.21
	MESD	40.17 $\pm$ 1.46	39.98 $\pm$ 1.51
Model merge+FT	Uniform	41.45 $\pm$ 1.53	41.17 $\pm$ 1.40
	Re-basin + Uni.	47.89 $\pm$ 1.32	48.13 $\pm$ 1.34
	Greedy Soup	45.90 $\pm$ 2.15	46.01 $\pm$ 2.21
	TIES	48.34 $\pm$ 1.07	48.80 $\pm$ 0.79
	Re-basin + TIES	48.56 $\pm$ 1.71	48.88 $\pm$ 1.36
	MOCHEE	50.85 $\pm$ 1.74	49.40 $\pm$ 1.94

via permutation before averaging and *TIES* [37] reduces destructive interference during merging. We also present *Re-basin* + *TIES*, which first aligns hidden units with *Re-basin* and then applies *TIES* for merging.

Evaluation and settings. All speech classifiers share the same architecture: a frozen wav2vec2-XLS-R-300m [69] encoder to support the cross-lingual setting, followed by a two-layer MLP head classifier. Given pretrained source-domain classifiers, we use the target-training instances to obtain a target classifier via model merging, and evaluate it on the target-test set. We report accuracy and macro-F1, averaged over five random seeds.

4.2. Target-domain Performance

Table 3 compares MOCHEE with existing model-merging baselines. Uniform averaging performs poorly, and permutation alignment alone (*Re-basin* + *Uni.*) provides only marginal gains, indicating that alignment without target-aware source weighting is insufficient. Greedy Soup and *TIES* incorporate more target-driven merging than uniform averaging, but their gains remain limited. Overall, MOCHEE achieves the best accuracy and macro-F1 across both target domains, showing the importance of jointly learning permutation-based alignment and target-aware merging weights from limited target labels.

Recall that model merging combines pretrained source models without directly fine-tuning them, offering computational efficiency. A natural question then arises: *how does performance change if we directly fine-tune the merged models on the target training set?* To answer this, Table 4 reports results on CAFÉ after fine-tuning each merged baseline on the target training set. We also include the performance of individually fine-tuned source-pretrained models for reference. We do not perform any additional fine-tuning for MOCHEE, as our soft-permutation alignment already supports effective target adaptation without updating model parameters. Notably, fine-tuning

Table 5: Ablation study on MOCHEE. Best results are in bold. Each component consistently improves the performance.

Target	CAFÉ		Oréau	
Method	Accuracy	Macro-F1	Accuracy	Macro-F1
w/o-perm	38.80 $\pm$ 3.50	34.09 $\pm$ 7.05	21.91 $\pm$ 2.74	15.33 $\pm$ 3.28
w/o- α	47.30 $\pm$ 2.19	43.45 $\pm$ 2.40	35.04 $\pm$ 1.73	34.51 $\pm$ 1.71
MOCHEE	50.85 $\pm$ 1.74	49.40 $\pm$ 1.94	35.30 $\pm$ 1.74	34.85 $\pm$ 2.17

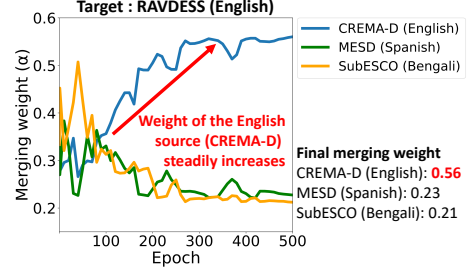


Figure 2: Evolution of source merging weights. When the target domain is English (RAVDESS), the weight assigned to the English source domain (CREMA-D) steadily increases during training, whereas the weights of linguistically distant sources (MESD and SubESCO) decrease.

improves several baselines, but MOCHEE still achieves the best performance, outperforming fine-tuned merged models.

4.3. Ablation Study

To evaluate the individual contributions of each module, we compare MOCHEE with two variants: MOCHEE-w/o-perm and MOCHEE-w/o- α in Table 5. MOCHEE-w/o-perm is MOCHEE without the representation alignment and MOCHEE-w/o- α is MOCHEE without updating the merging scores. Note that removing permutation alignment causes a large drop, while removing α yields a smaller but consistent degradation. Overall, MOCHEE achieves the highest accuracy and macro-F1, indicating that both modules consistently contribute to performance.

4.4. Merging Weights of Source Domains

To analyze the evolution of MOCHEE’s merging weights α , we construct a setting where English appears in both the source and target domains by selecting RAVDESS (English) as the target and including CREMA-D (English) among the sources. This allows us to directly observe whether MOCHEE assigns larger weights to sources that are linguistically similar to the target.

As shown in Figure 2, MOCHEE progressively increases the weight of the English source (CREMA-D). In contrast, linguistically distant sources, which are MESD (Spanish) and SubESCO (Bengali), receive lower weights. The merging weights stabilize toward the end of training, demonstrating that they are learned in a stable and target-aware manner.

5. Conclusion

We propose MOCHEE, a target-centric model merging method for source-free speech classification under scarce target supervision. MOCHEE jointly learns target-guided alignment and source importance via a meta-learning objective that selects merging parameters to improve validation generalization rather than fit the limited target training set. Experiments on real-world cross-corpus speech benchmarks show that MOCHEE consistently outperforms existing merging baselines.

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7. Generative AI Use Disclosure

We use generative AI tools only for language editing and polishing (e.g., improving grammar and clarity). All technical content, claims, experimental design, results, and interpretations are produced, verified, and approved by the authors, who take full responsibility for the paper.

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